

Crossover Designs

PUBH 526

Design and Analysis of Randomized Trials in Public Health

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Crossover Design

Soluble corn fiber increases bone calcium retention in postmenopausal women in a dose-dependent manner: a randomized crossover trial¹

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- American Journal of Clinical Nutrition 2016;104:837–43.

ABSTRACT

Background: Dietary soluble corn fiber (SCF) significantly improves calcium absorption in adolescents and the bone strength and architecture in rodent models.

Objective: In this study, we aimed to determine the skeletal benefits of SCF in postmenopausal women.

Design: We used our novel technology of determining bone calcium retention by following the urinary appearance of ^{41}Ca , a rare long-lived radioisotope, from prelabeled bone to rapidly and sensitively evaluate the effectiveness of SCF in reducing bone loss. A randomized-order, crossover, double-blinded trial was performed in 14 healthy postmenopausal women to compare doses of 0, 10, and 20 g fiber from SCF/d for 50 d.

Results: A dose-response effect was shown with 10 and 20 g fiber from SCF/d, whereby bone calcium retention was improved by 4.8% ($P < 0.05$) and 7% ($P < 0.04$), respectively. The bone turnover biomarkers *N*-terminal telopeptide and osteocalcin were not changed by the interventions; however, a significant increase in bone-specific alkaline phosphatase, which is a bone-formation marker, was detected between 0 and 20 g fiber from SCF/d (8%; $P = 0.035$).

Conclusion: Daily SCF consumption significantly increased bone calcium retention in postmenopausal women, which improved the bone calcium balance by an estimated 50 mg/d. This study was registered at clinicaltrials.gov as NCT02416947. *Am J Clin Nutr* 2016;104:837–43.

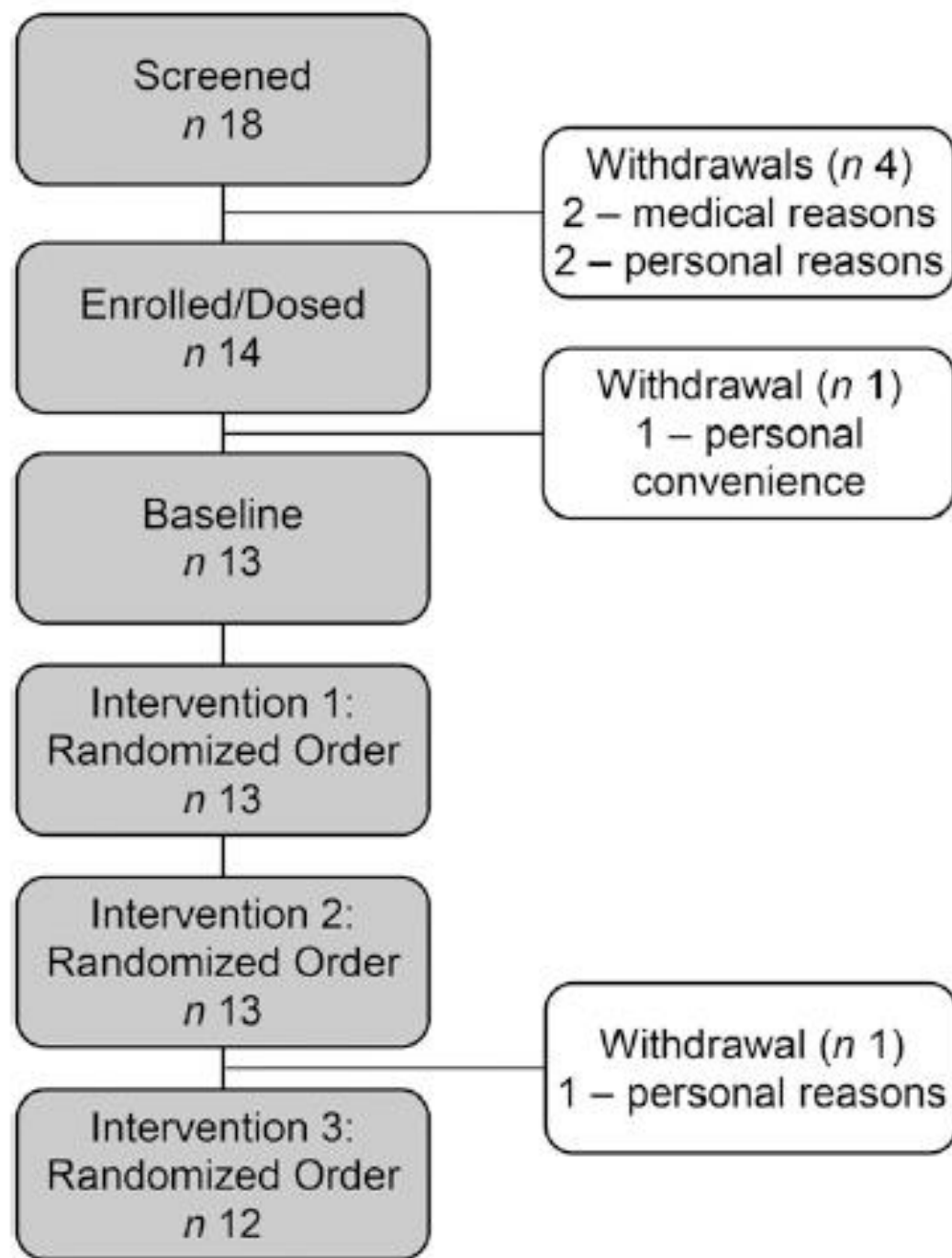


FIGURE 1 Participant flow through the study.

Hall et al. 2019

Crossover Design

- Suppose you have two treatments, A and B
- How many Latin Squares could you make?
 - How many ways could you order two treatments over time?

Latin Squares																				
	I		II		III		IV		V		VI		VII		VIII		IX		X	
Subject	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Period 1	A	B	B	A	B	A	A	B	A	B	B	A	A	B	A	B	A	B	A	B
Period 2	B	A	A	B	A	B	B	A	B	A	A	B	B	A	B	A	B	A	B	A

- How about 3 treatments? 4 treatments?
- With 2 treatments, what are the possible covariance structures?

Randomization

- Want to evaluate 3 treatments in a crossover design
- Power calculations say you need 22 participants
- In what order would you assign the 3 treatments to each participant?

Crossover Design

- A commonly-used within-subject design
- Used in drug comparisons, physiology experiments, consumer acceptance studies...
- Consider s subjects as blocks
- Each subject undergoes p treatments over p periods
- Can consider incomplete block structure:
 - # periods $<$ # treatments
 - Too many treatments for one subject to handle
- Have a break between periods to remove residual effect (a.k.a. carryover effect)

Crossover Design

- Used because one anticipates high level of variability between subjects → block on subject to remove that variability
- Subject (S_k) is “serving as its own control”
- Commonly used for 2, 3, or 4 periods
- Period (P_i) is typically considered a blocking factor too

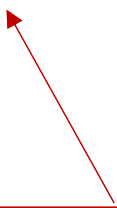
Model

$$y_{ijk} = \mu + P_i + \tau_j + S_k + \epsilon_{ijk}$$

- ϵ_{ijk} may be correlated
- Subjects often random effects
- Period may be significant due to
 - Biological reasons
 - Y is a more subjective measure
- Note we're doing a repeated measures analysis here

SAS (participants random)

```
proc mixed covtest cl data=cow;  
  class cow trt period;  
  model resp=trt period / ddfm=kr outp=diag;  
  repeated period / subject=cow type=cs r=1;  
  lsmeans trt / diff;  
run;
```



Other covariance
structures may be
explored

Pros and Cons of Crossover Designs

- Advantage
 - Participants serve as their own control to reduce error
- Disadvantages
 - Participants must be available for longer periods of time
 - Carryover issue: Are subjects the “same” at the start of each period?
 - Possible to model residual effects

Example: “a1c.sas”

- You are testing the effect of 3 dietary interventions on hemoglobin A1C in a crossover design with 15 participants